



Introduction to Cybersecurity

Ass. Prof. Mohammed Abdel Hamid

Faculty of Computers & Information

Luxor University

Computer Science Dept.

Lecture 1

Lecture Rules



Course contents

- 1. Introduction to Computers.**
- 2. Cybersecurity basics.**
- 3. Cyberattacks and Vulnerabilities.**
- 4. Recent Cyberattacks and their impact.**
- 5. Cryptography basics Part I.**
- 6. Cryptography basics Part II.**
- 7. Midterm Exam.**
- 8. Computer Malware and Classification.**

Course contents

- 9. Countermeasures for Cyberattacks Part I.**
- 10. Countermeasures for Cyberattacks Part II.**
- 11. Network security basics (Wireless Networks).**
- 12. 12th Week Exam.**
- 13. Cybersecurity Risk Management basics.**
- 14. E-Commerce Security & Cybersecurity standards.**
- 15. Revision.**
- 16. Final Exam.**

Introduction to Computers

Lecture Outline

- **Introduction.**
- **History of Computers.**
- **Components of Modern Computer.**
- **Future Machines.**
- **Computer Networks.**
- **What is Internet?**

Introduction

- Computer is an **electronic** device that takes the **raw data** through the **input components** and **processes** that into **meaningful information** for **efficient use**
- A **complete unit** of modern computer system consists of two major categories – referred to as the **software components** and the **hardware components**.



Introduction

- The **software components** consist of the following:
 - **Firmware**
 - **Operating system** or OS software
 - **Device drivers**
 - **Application software**
 - **Utility software.**
- While the **hardware components** consist of the following:
 - **Central** processing unit (**CPU**).
 - **Data** storage.
 - **Input** devices.
 - **Output** devices (input and output devices are often together called “**I/O devices**”).

History of Computers

1. First Generation 1940-1956

- The **first-generation** computers were based on the **vacuum tubes**. The name of this machine was **Electronic Numerical Integrator and Computer (ENIAC)**.

2. Second Generation 1956-1964

- The **second-generation** computers were based on the **transistor technology**. The first commercial computer named as Universal Automatic Computer (**UNIVAC 1**)

3. Third Generation 1964-1971

- The third-generation computers were based on the **integrated circuits or ICs**.

History of Computers

4. Fourth Generation 1971-2020

- These computers were mostly based on the **very large-scale integration (VLSI) microprocessors**
- For Ex. Personal computers (PCs), laptops, personal digital assistants (PDAs), and other computer machines were introduced under this generation of computers .
- Windows, Linux, UNIX, and Macintosh operating systems.

5. Fifth Generation 2010-present

- The fifth-generation computers are those machines that are powered by the **artificial intelligence (AI)** and **ultra-large-scale integration (ULSI)** .

History of Computers

First Generation (1940 to 1956)



Vacuum tubes

Second Generation (1956 to 1964)



Transistor technology.

Third Generation (1964 to 1971)



**Integrated circuits
or ICs.**

Fourth Generation (1971 to present)



**(VLSI)
microprocessors**

Fifth Generation (Present & beyond)



AI and ULSI



Components of Modern Computer

1. Input Devices

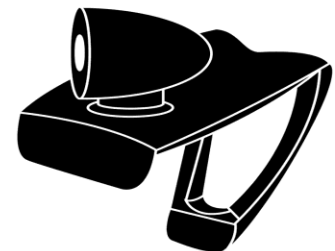
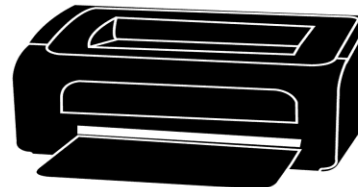
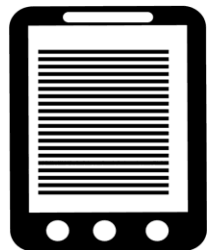
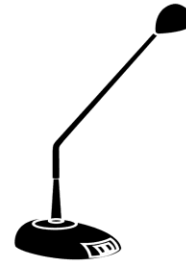
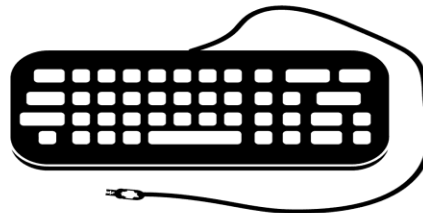
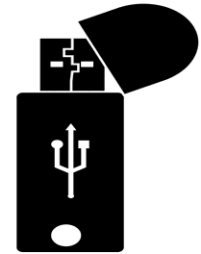
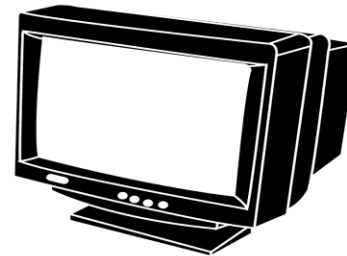
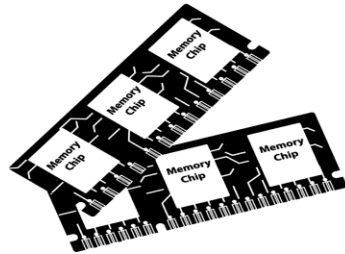
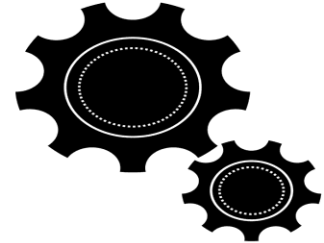
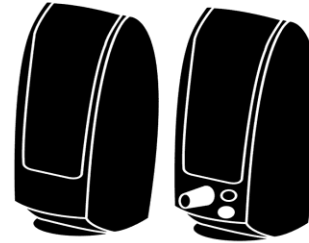
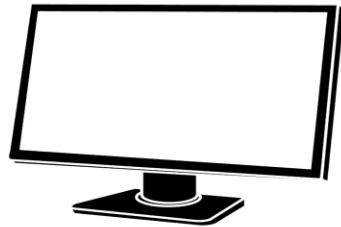
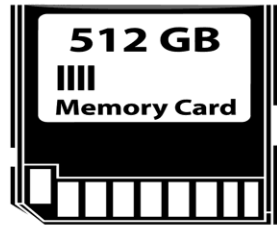
- Those components that are connected to the computer to **insert raw** data in **different formats** for **processing** purpose.
- The major input **devices** for our **modern** computers include the **following**:
 - Keyboard
 - Barcode reader
 - Electronic pen
 - Wireless antenna
 - Scanner
 - Microphone
 - Camera
 - Joystick
 - USB port

Components of Modern Computer

2. Output Devices

- Those **components** that are **connected** to the **computer** for **producing meaningful information** processed from the raw data.
- The major output devices of a modern computer include the following:
 - Monitor
 - Printer
 - Speaker
 - Headphone
 - Projector
 - Plotter

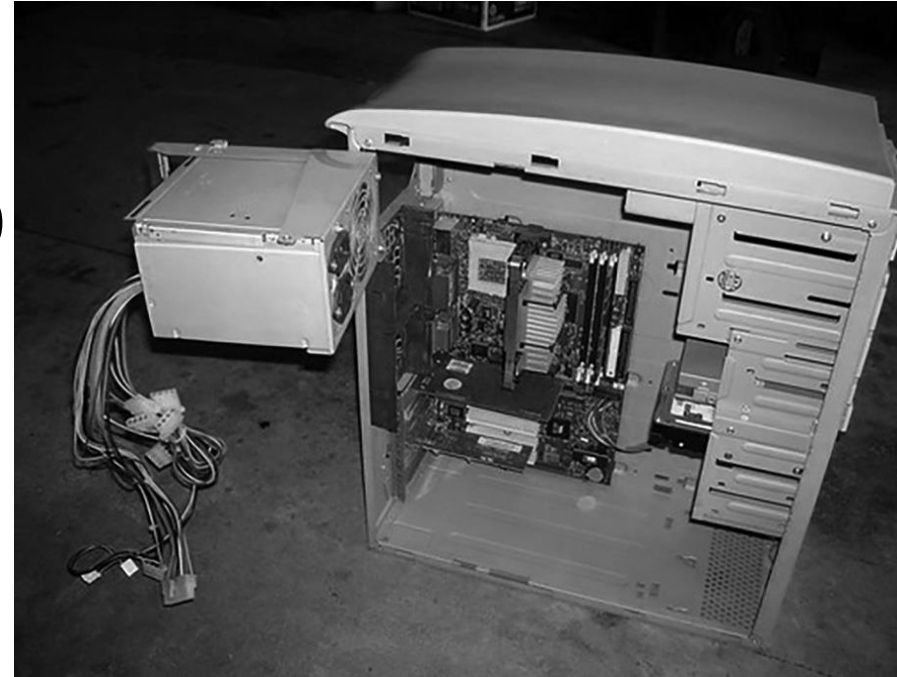
Components of Modern Computer



Components of Modern Computer

3. Central Processing Unit (CPU)

- The **CPU** is the **brain** of computer that **controls** the entire **functionalities** of the **machine** to generate and **display** the **meaningful** information or **saves** it on the **data storage**.
- The major components of a CPU include the following:
 - Arithmetic logic unit (**ALU**)
 - Random access memory (**RAM**)
 - Read-only memory (**ROM**)
 - Control unit (**CU**)



Components of Modern Computer

4. Storage Components

- The **data storage** bank to keep the **data** saved on the **secondary location**, where the **data** can be easily **accessed** and **managed**
- The storage components include the following:

- Hard disk drive (**HDD**)
- Solid State drive (**SSD**)
- Flash disk drive (**FDD**)
- Optical disk drive (**ODD**)
- **Magnetic tapes**

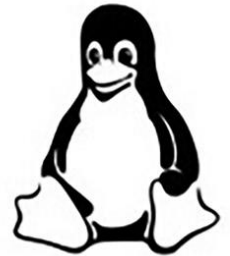


Components of Modern Computer

5. Software Components

- All **hardware components** are **deaf** without the **power** of **software components** of a computer.
- The major **software components** of a computer include the following:

- **OS software.**
- **Component drivers.**
- **Application** software.
- **Utility** software.
- **Firmware.**



Future Machines

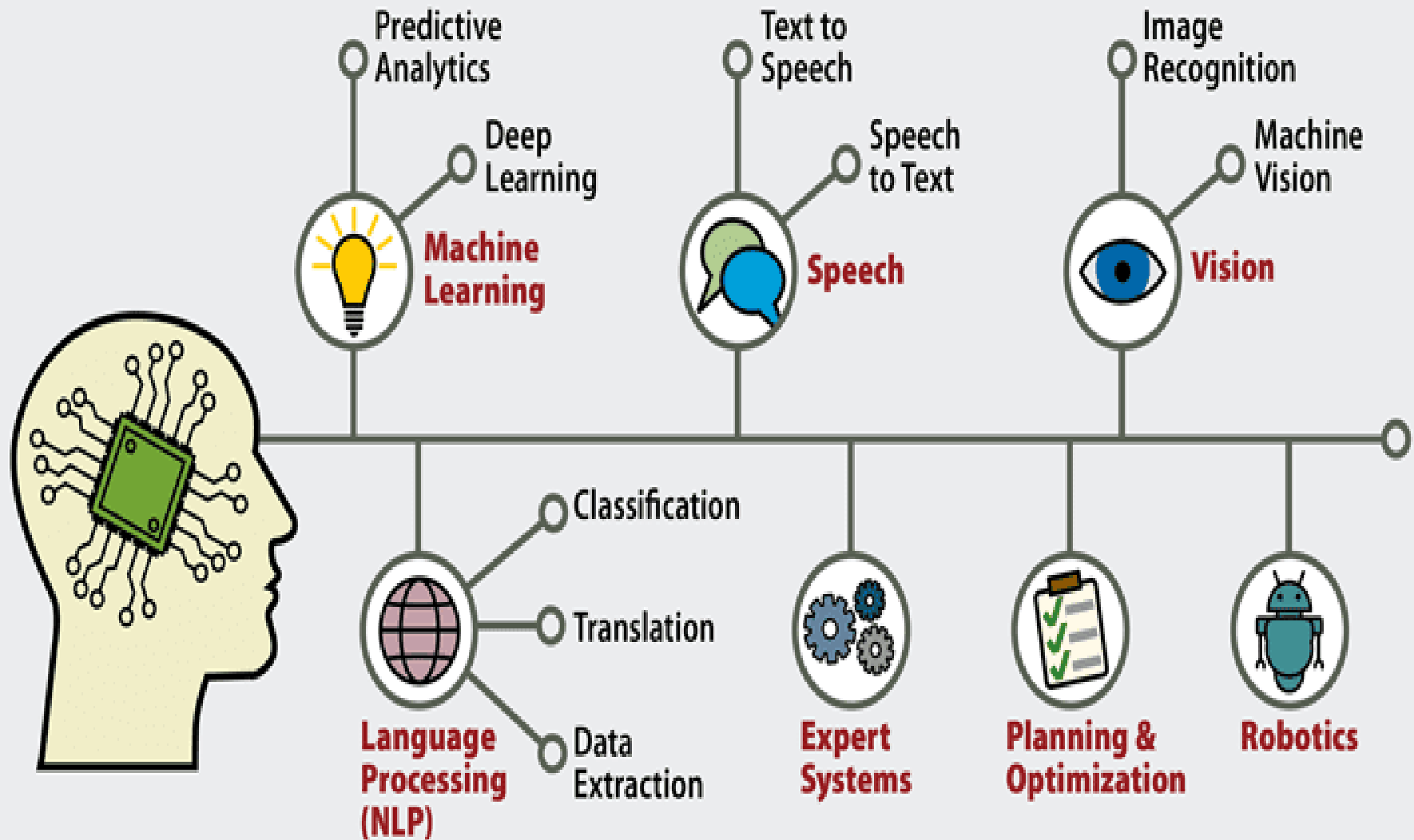
1. Artificial Intelligence

- The **future** of **computers** looks **even smarter** and more **exciting** with **AI technologies** such as **augmented reality** and **virtual reality** (AR/VR), **voice recognition**, **text interpretation**, **image processing** and **motion detection**.

2. Quantum Computing

- Quantum computers could be used to **improve the secure sharing** of **information**. Or to **improve radars** and their ability to detect **aircraft**. Another area where quantum computing is expected to help is the **environment** and **keeping water clean** with **chemical sensors**.

Artificial Intelligence



TEN QUANTUM COMPUTING APPLICATIONS

CYBERSECURITY



BETTER BATTERIES



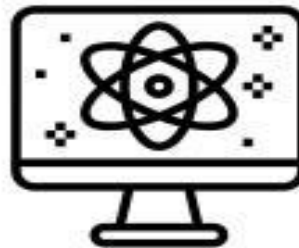
DRUG DEVELOPMENT



WEATHER FORECASTING



ARTIFICIAL INTELLIGENCE



TRAFFIC OPTIMISATION



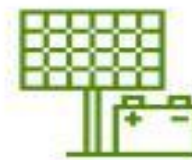
ELECTRONIC MATERIAL DISCOVERY



FINANCIAL MODELING



CLEANER FERTILISATION



SOLAR CAPTURE

Future Machines

3. Nanotechnology

- The **size of the hardware**, especially in **electronics** and **computer technology**, is **decreasing exponentially**.

4. Parallel Processing

- The distribution of **computer processing** in **multiple components** at the same time to **increase** the **processing power** of a **computer**.
- For example: The **virtualization**, **cloud computing**, and **distributed processing systems**.



NANO DEVICES



**OPTICAL
ENGINEERING**



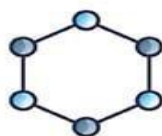
**DEFENSE AND
SECURITY**



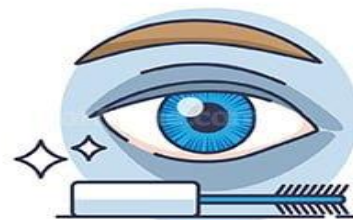
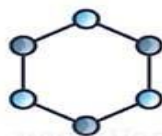
**BIO
ENGINEERING**



**MEDICINE AND
DRUGS**



NANOTECHNOLOGY APPLICATIONS



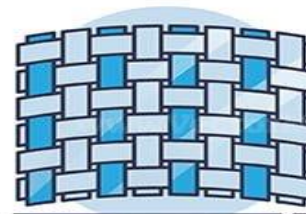
COSMETICS



ENERGY

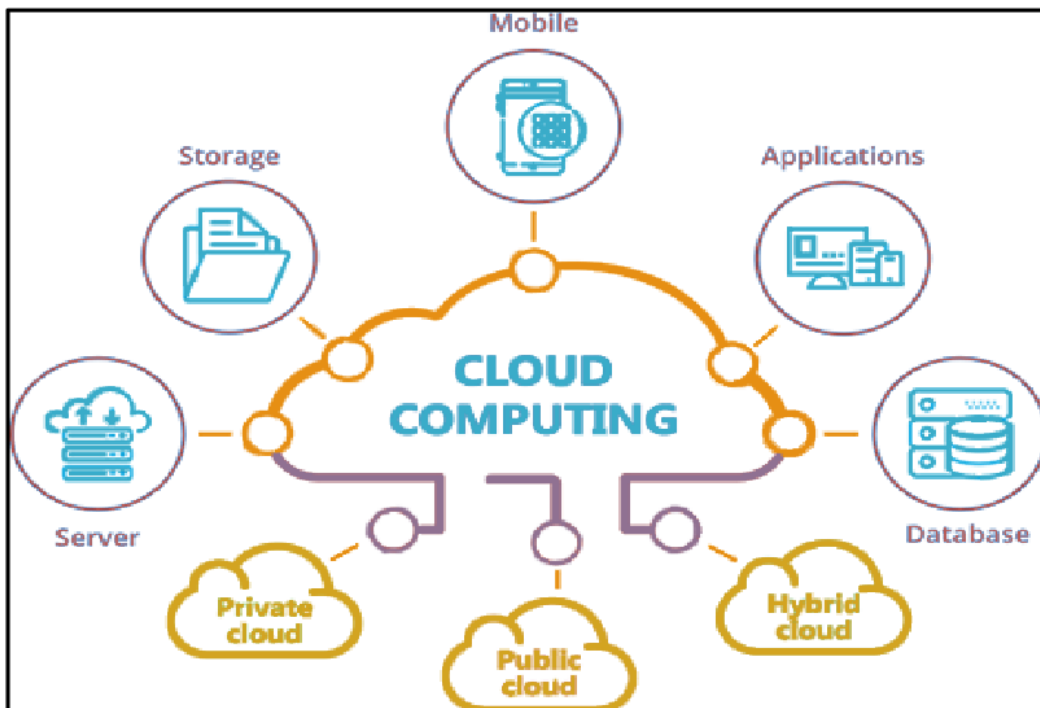
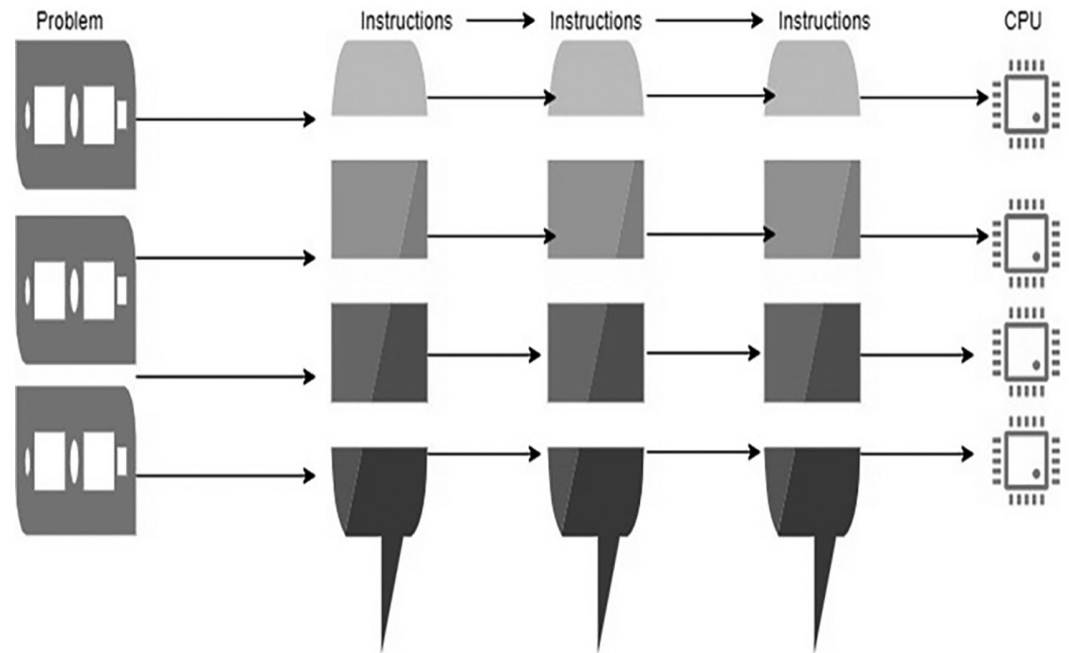


NANOBIO-TECHNOLOGY



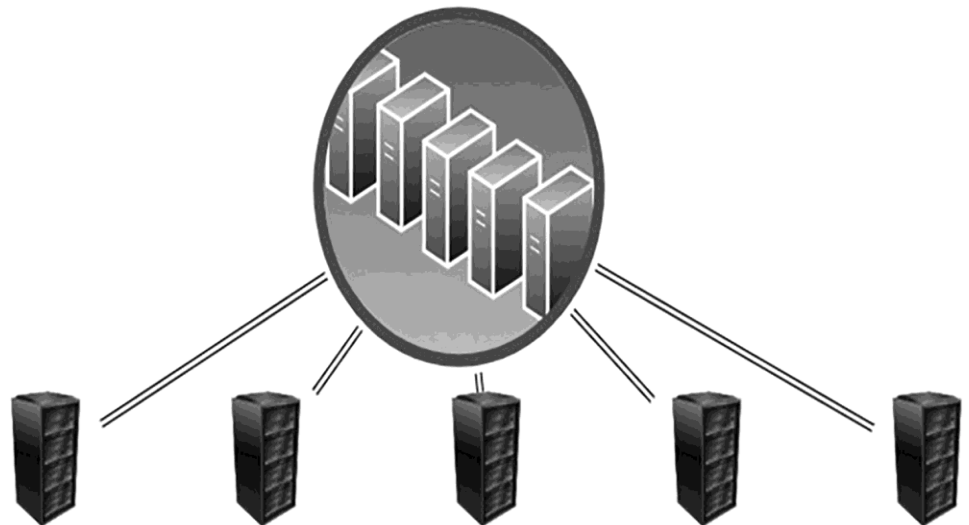
NANO FABRICS

Parallel Processing



Computer Networks

- **Computer network** is the **combination** of **two** or **more computers** connected with each other to **share their resources**, establish communication, and **exchange** data **electronically** based on **some rules** and **protocols**.
- A computer network consists of **networking devices** (e.g., switches, hubs, routers), **computer hosts**, **wired media** (copper or fiber cables), **wireless media**, **connectors**, and **software protocols**.



Computer Networks

- **Multiple computers** are connected with each other **through different configurations (physical and logical)** known as **network topologies**.

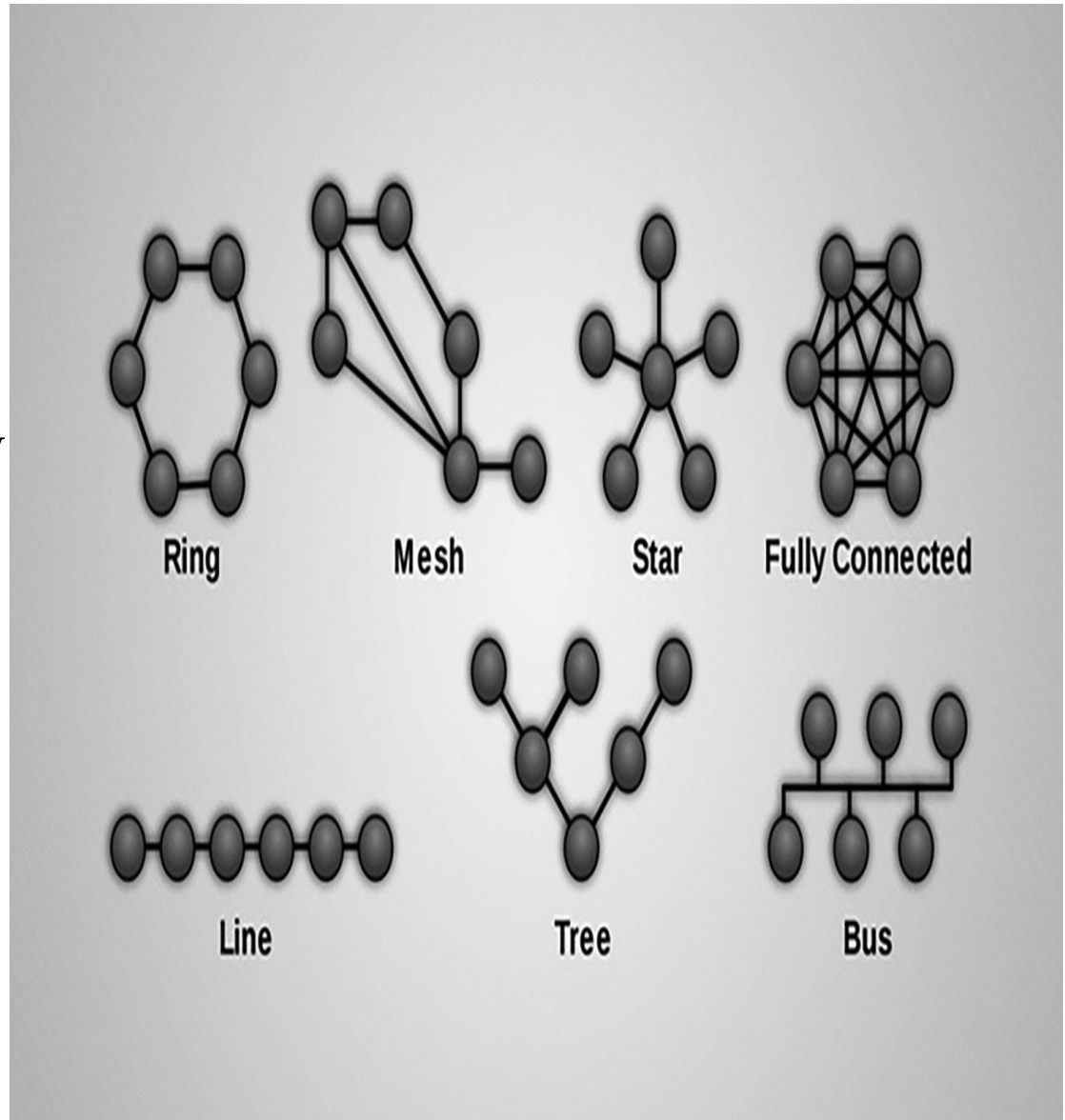
Network Topologies

- **Network** topology is a type of **computer connectivity** in which the **computers** or **devices** are **connected** with **each** other to **form a network**.
- These **topologies** are normally **physical references** of **network connectivity**.

Computer Networks

Network Topologies

- **Star** network topology
- **Bus** network topology
- **Ring** network topology
- **Mesh** network topology
- **Tree** topology
- **Combination** of two or more topologies.



Computer Networks

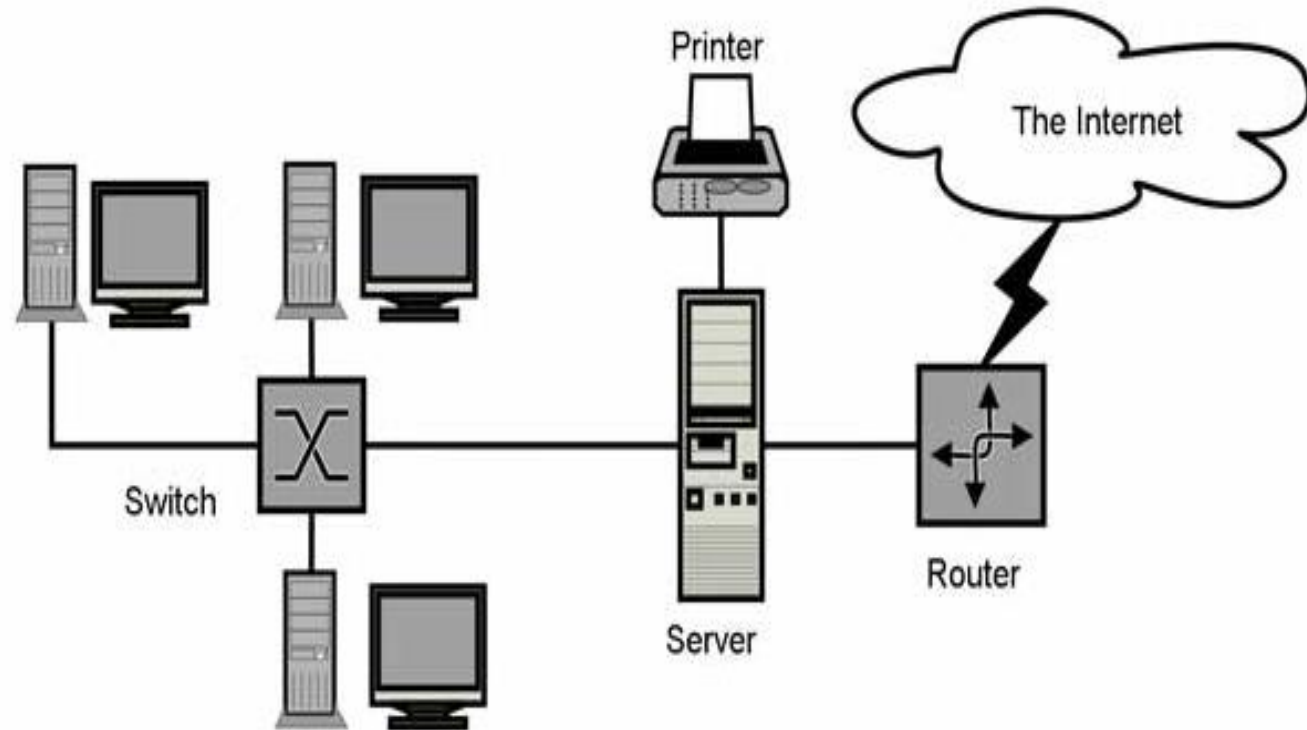
Major Network Protocols of OSI (Open Systems Interconnection)

- **Ethernet** protocol for **physical and data link layer**.
- **TCP/SPX** (Transmission Control Protocol/Sequenced Packet Exchange) at **Transport layer**.
- **IP** (Internet Protocol), **ICMP**, **IGMP** at **Network layer**.
- **HTTP/FTP** (Hypertext Transfer Protocol/File Transfer Protocol) **upper OSI layers (Application layer)**.
- **DNS** (Domain Name System), **SMTP** (Simple Mail Transfer Protocol) **upper OSI layers. (Application layer)**.

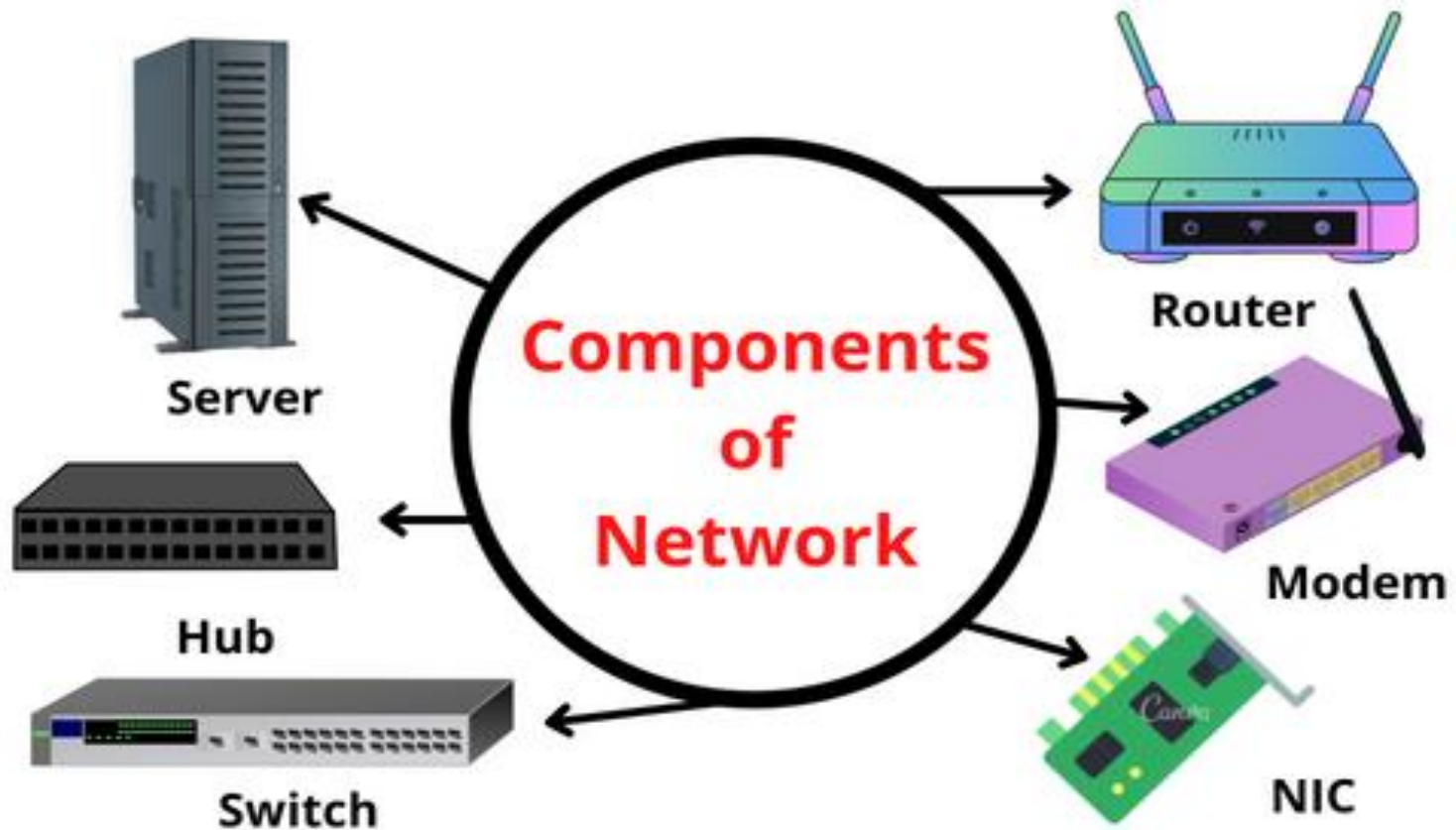
Computer Networks

Components of Computer Networks

- Hosts
- Routers
- Switches
- Hubs
- Firewalls
- Servers
- Cables/links
- Protocols



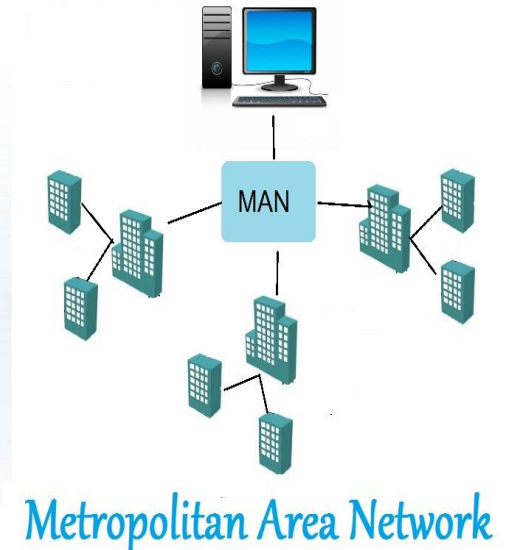
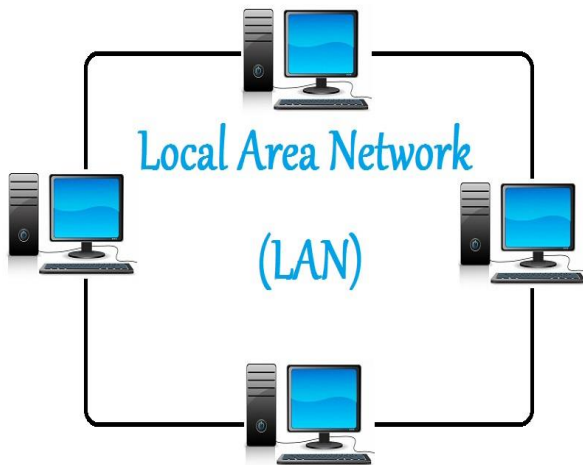
Computer Networks



Computer Networks

Types of Networks

- These networks are classified in terms of their **coverage area**. They can be **wired**, **wireless**, or even a **combination** of **both**.
1. Local area network (**LAN**)
 2. Metropolitan area network (**MAN**)
 3. Wide area network (**WAN**)



What is Internet?

- The **Internet** is the **global network** of the **networks** spread all **over** the **world**.
- This **network** uses the **IP address** as the **identification address** of any **node** in the **Internet**.
- The **complete communication** stack for the **transmission** of data across the **network** is based on the **TCP/IP**.

What is Internet?

- **The major services offers by Internet include the following:**
- Voice calls
- Video calls
- Chats
- Fax
- Email
- Social media
- Instant messaging
- Financial services
- Bill payments
- Online shopping
- TV streaming

What is Internet?

- The **major services** offers by **Internet** include the following:
- Live movies
- Gaming
- File sharing
- Data transfer
- e-Banking
- Data storage
- e-Marketing
- Remote healthcare
- Remote working
- e-Learning

thanks for
listening!